

$$\begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

(27)

$$A = \begin{bmatrix} 9 & 15 \\ 15 & 50 \end{bmatrix}$$

$$A = F F^T$$

$$F = \begin{bmatrix} l_{11} & l_{12} \\ l_{21} & l_{22} \end{bmatrix}$$

$$A = L L^T$$

$$L = \begin{bmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{bmatrix}$$

$$L^T = \begin{bmatrix} l_{11} & l_{21} \\ 0 & l_{22} \end{bmatrix}$$

$$L L^T = \begin{bmatrix} l_{11}^2 & l_{11} l_{21} \\ l_{11} l_{21} & l_{21}^2 + l_{22}^2 \end{bmatrix} = \begin{bmatrix} 9 & 15 \\ 15 & 50 \end{bmatrix}$$

$$l_{11}^2 = 9 \rightarrow |l_{11}| = 3$$

$$l_{11} l_{21} = 15 \rightarrow |l_{21}| = \frac{15}{3} \Rightarrow |l_{21}| = 5$$

$$l_{21}^2 + l_{22}^2 = 50$$

$$25 + l_{22}^2 = 50$$

$$l_{22}^2 = 25 \rightarrow |l_{22}| = 5$$

$$L = \begin{bmatrix} 3 & 0 \\ 5 & 5 \end{bmatrix}$$

$$L^T = \begin{bmatrix} 3 & 5 \\ 0 & 5 \end{bmatrix}$$